Magnetically driven ferrofluid channel flow

Formulation, implementation and verification — with plain-language intuition

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Magnetically driven ferrofluid channel flow: formulation, implementation and verification

A phase-field MRT lattice-Boltzmann solver for a planar channel, driven by a magnetic field that both pushes the fluid and changes its viscosity. This version explains each idea in plain language first, then gives the full mathematics. All quantities are in lattice units unless stated otherwise.

How to read this document

Every section starts with a short paragraph in plain words — the picture you should carry in your head — followed by the equations and the exact numbers that make it precise. The plain paragraph never replaces the math; it tells you what the math is *for*. If you already know the field, skim the intuition and read the equations; if you are new to it, read the intuition twice and let the equations catch up.

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1. Objective and configuration

The idea. In the last phase, we pushed the fluid down the channel with a plain mechanical shove (a body force), and used an oscillating magnetic field only to change how thick the fluid behaves — its viscosity. Now we take the mechanical shove away entirely. The *same* magnetic field has to do both jobs: push the fluid **and** tune its viscosity. The trick that makes the field push is to make it stronger at one end of the channel than the other, so the magnetic particles get pulled toward the strong end, dragging the fluid with them.

The setup. The earlier phase reproduced Bacri's rotational-viscosity curves in a channel driven by a constant body force F_b, with a uniform oscillating field modulating the viscosity. This phase removes the body force, $F_b = 0$, and drives the flow with a spatially graded magnetic field, so that both the pumping and the viscosity modulation are magnetic.

Geometry: a 2D channel, periodic in x, no-slip walls at $y = 0$ and $y = H$, with $n_X = n_Y = 50$. Lattice nodes sit at $x_i = i$ and $y_j = j + \frac{1}{2}$, so the walls are halfway links, consistent with bounce-back. The field centre is $x_c = (n_X - 1)/2$ and $y_c = n_Y/2$.

Fluid parameters: $\rho_h = 1.975$, $\eta_h = 0.05$, $\phi_p = 0.2$, $\tau_B = 20$, $H_{ref} = 1.73$, $\mu_0 = 1$, $c_s^2 = 1/3$.

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2. The applied field

2.1 What Maxwell allows

The idea. You might think the simplest way to make a field that grows along the channel is $\mathbf{H} = (H_0 + kx,\ 0)$ — pointing along x, getting stronger as x grows. Nature forbids it. A

magnetic field in empty space obeys two rules: it has no sources ($\nabla \cdot \mathbf{H} = 0$, field lines never begin or end), and it has no swirl ($\nabla \times \mathbf{H} = 0$, no currents here). The naive field breaks the first rule. The fix Nature insists on is that if the field grows along x, it *must* tilt a little in y to keep the field lines from piling up. That small sideways tilt, $H_y = -k(y - y_c)$, is not a modelling choice — it is forced, and later it turns out to matter physically.

The mathematics. The applied field is produced by external currents, so inside the channel

$$\nabla \times \mathbf{H} = 0, \qquad \nabla \cdot \mathbf{H} = 0.$$

The first condition gives $\mathbf{H} = -\nabla\psi$, and the second makes ψ harmonic, $\nabla^2\psi = 0$. This immediately rules out the naive choice $\mathbf{H} = (H_0 + kx,\ 0)$, since its divergence is $k \neq 0$. It also cannot be written with a vector potential: with $B_y = -\partial_x A = 0$ we get $A = A(y)$, so $B_x = A'(y)$ cannot depend on x.

Expanding ψ to second order about the channel centre, with $X = x - x_c$ and $Y = y - y_c$,

$$\psi = -\alpha X - \beta Y - \tfrac{1}{2}\left(aX^2 + 2bXY + cY^2\right),$$

$$H_{x,app} = \beta + bX + cY, \qquad H_{y,app} = -(\alpha + aX + bY).$$

Two conditions fix the coefficients:

- harmonic: $a + c = 0$;
- symmetric about the centreline, i.e. H_x even and H_y odd in Y: $c = 0$, hence $a = 0$, and $\alpha = 0$.

Writing $\beta = H_d$ and $b = k$, and adding the uniform oscillating field:

$$\boxed{\mathbf{H}_{app} = \Big(H_d + H_0 \cos\omega t + k\,(x - x_c),\quad -k\,(y - y_c)\Big)}$$

$$A_{app} = \mu_0\left[H_d + H_0 \cos\omega t + k(x - x_c)\right](y - y_c), \qquad \nabla^2 A_{app} = 0.$$

The transverse component $H_y = -k(y - y_c)$ is **not optional**. It is the price of $\partial_x H_x = k$ under $\nabla \cdot \mathbf{H} = 0$. It turns out to carry real physics (Section 6.3).

2.2 Why the uniform part H_d cannot be zero

The idea. The magnetic push points from weak field toward strong field — the particles want to sit where the field is biggest. If you turn off the steady part H_d and keep only the gradient, the field strength forms a bowl with its bottom at the centre of the channel: zero in the middle, growing in every direction. Particles then get shoved *outward* on all sides, and by symmetry the net push down the channel is zero. To get a one-directional push you need a steady background field H_d that is strong enough that the field never dips to

zero anywhere in the channel — so the "strong end" is always genuinely the downstream end.

The mathematics. With **M** aligned with **H** and $\nabla \times \mathbf{H} = 0$,

$$(\mathbf{H}\cdot\nabla)\mathbf{H} = \nabla\big(\tfrac{1}{2}|\mathbf{H}|^2\big) \quad\Longrightarrow\quad \mathbf{F}_K = \mu_0(\mathbf{M}\cdot\nabla)\mathbf{H} = \mu_0 M\,\nabla|\mathbf{H}|.$$

If $H_d = 0$ (and no oscillating part), $\mathbf{H} = k(X, -Y)$ is a pure quadrupole with $|\mathbf{H}| = k\sqrt{X^2+Y^2}$, which vanishes at the channel centre. The force then points radially away from that null, changes sign at $X = 0$, and gives no net drive. Two requirements follow:

$H_d > \dfrac{kL}{2}$ (no null inside the channel), $\qquad H_d \gg kL$ (needed by the seam closure, Section 4.3).

When both hold, $F_{K,x} \approx \mu_0 k\, M(H_d)$ is nearly uniform and plays the role of the old body force, while $F_{K,y}/F_{K,x} \approx k(y - y_c)/H_d \lesssim 10^{-4}$.

2.3 The alternative with $F_{K,y} \equiv 0$, and why it was not used

The idea. The forced sideways tilt bothered us, so we asked: is there a field whose push is purely along x, with no sideways component at all? Mathematically there is exactly one such family — a field that grows *exponentially* along x while its direction rotates as you cross the channel. We did not use it. Our simpler field (linear growth) is just this exotic field with the tiny curvature thrown away; the difference is one part in ten million. And the exotic field breaks the neat bookkeeping we use at the channel's periodic seam. So we kept the simple field and dealt with its sideways tilt honestly, which, as it happens, is where a genuine new piece of physics was hiding.

The mathematics. For aligned **M**, $F_{K,y} = 0$ requires $|\mathbf{H}|$ to depend on x only. In 2D, $\mathcal{F}(z) = H_x - iH_y$ is analytic, $\ln|\mathcal{F}|$ is harmonic, and if it depends on x alone it must be linear in x. The unique family is therefore

$$\mathbf{H} = H_A e^{a(x-x_0)}\big(\cos a(y-y_0),\ -\sin a(y-y_0)\big), \qquad A = \frac{\mu_0 H_A}{a} e^{a(x-x_0)} \sin a(y-y_0),$$

checked symbolically for $\nabla\cdot\mathbf{H} = 0$, $\nabla\times\mathbf{H} = 0$, $\nabla^2 A = 0$, and $(\mathbf{M}\cdot\nabla)\mathbf{H} = m\,a\,\sigma\,\hat{x}$ exactly.

It was rejected because (i) the linear field is its first-order truncation with $k = aH_A$, and the two differ by $O((kH/H_d)^2) \sim 10^{-7}$; (ii) its seam jump is multiplicative, $A(x+L) = e^{aL}A(x)$, so the exact lattice jump of Section 4.3 no longer applies; and (iii) $F_{K,y}$ in the linear field is a pure y-gradient that changes only the pressure (Section 5).

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3. Governing equations

The idea. Three physical stories run at once, and the solver keeps a separate "population lattice" for each. First, the fluid moves: ordinary Navier-Stokes, but with two extra

magnetic pushes added to the usual pressure and friction. Second, the magnetic field has to arrange itself consistently given where the particles are: a magnetostatics problem solved through a vector potential A. Third, each blob of magnetic particles has an orientation that lags behind the field and gets twisted by the flow: a little rotational relaxation equation for the magnetization $\mathbf{M}$. These three talk to each other every timestep, which is the whole difficulty and the whole point.

3.1 Flow

$$\nabla \cdot \mathbf{u} = 0, \qquad \rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \nabla \cdot \left[\eta(\nabla \mathbf{u} + \nabla \mathbf{u}^T)\right] + \underbrace{\mu_0(\mathbf{M} \cdot \nabla)\mathbf{H}}_{\text{Kelvin}} + \underbrace{\tfrac{1}{2}\nabla \times (\mu_0 \mathbf{M} \times \mathbf{H})}_{\text{antisymmetric stress}} + \mathbf{F}_s,$$

with $\mathbf{F}_b = 0$. The two magnetic terms are the whole story of this phase: the **Kelvin force** is the pull toward strong field (the pump), and the **antisymmetric stress** is the force from particles being twisted out of alignment (this is what becomes rotational viscosity). The f-lattice is the pressure-based MRT scheme; the g-lattice is the conservative Allen-Cahn phase field ($\phi \equiv 1$ here, single phase).

3.2 Magnetostatics

The idea. We never solve for $\mathbf{H}$ directly. We solve for a helper field A (the vector potential) from which $\mathbf{B}$, and then $\mathbf{H}$, follow by taking derivatives. The reason is bookkeeping: writing $\mathbf{B}$ as the curl of A makes "no magnetic sources" ($\nabla \cdot \mathbf{B} = 0$) true automatically, for free, no matter what. That saves us from enforcing it by hand.

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = (\partial_y A,\ -\partial_x A) \ \Rightarrow\ \nabla \cdot \mathbf{B} = 0 \text{ identically},$$

$$\nabla^2 A = -\mu_0 \left(\nabla \times \mathbf{M}\right)_z, \qquad \mathbf{H} = \frac{1}{\mu_0}\left(\partial_y A,\ -\partial_x A\right) - \mathbf{M}.$$

The h-lattice solves the first of these in pseudo-time, $\partial_s A = \epsilon \nabla^2 A + \epsilon\left[\mu_0(\nabla \times \mathbf{M})_z + r\right]$, with $\epsilon = 9.2$, `NSUB_H` $= 64$ sub-steps per physical step, and r the compatibility constant. (Pseudo-time means we relax A to its steady solution with a fake diffusion, many sub-steps per real timestep, because the magnetic field settles far faster than the fluid moves.)

3.3 Magnetization: MRSh effective field

The idea. A magnetic nanoparticle in a field wants to align with it, but it takes a characteristic time τ_B (the Brownian relaxation time) to swing around. In a steady field it points along the field, with a magnitude set by the Langevin function L (weak field: proportional to field; strong field: saturated). In an *oscillating* field it can never quite keep up — it lags. And in a shear flow, the surrounding fluid physically rotates the particle, tilting it away from the field. The "effective field" model (MRSh, Martsenyuk-Raikher-Shliomis) packages all this into an equation for a vector $\boldsymbol{\xi}_e$ that tracks where the magnetization is actually pointing, as opposed to where the field is.

$$\mathbf{M} = \phi M_s L(\xi_e)\,\hat{\mathbf{e}}, \qquad \hat{\mathbf{e}} = \boldsymbol{\xi}_e/\xi_e, \qquad L(x) = \coth x - 1/x,$$

$$\frac{d\boldsymbol{\xi}_e}{dt} = \boldsymbol{\Omega}\times\boldsymbol{\xi}_e - \frac{1}{\tau_B}\frac{L(\xi_e)}{L'(\xi_e)}\left(1-\frac{\xi_\parallel}{\xi_e}\right)\hat{\mathbf{e}} + \frac{1}{2\tau_B}\left(\frac{\xi_e}{L(\xi_e)}-1\right)\boldsymbol{\xi}_\perp.$$

The three terms are, in order: the flow spinning the magnetization ($\boldsymbol{\Omega}\times$), the magnitude relaxing toward its equilibrium value, and the direction relaxing back toward the field. Here $\boldsymbol{\xi} = (\xi_0/H_{ref})\mathbf{H}$ and $\boldsymbol{\Omega} = \frac{1}{2}(\nabla\times\mathbf{u})_z\hat{z}$. Integrated with RK4, `NSUB_M` $= 8$ sub-steps per lattice step, $\mathbf{H}$ interpolated linearly across the step. The parameter mapping is

$$M_s = \frac{3\eta_h\phi_p\,\xi_0}{\mu_0\tau_B H_{ref}}, \qquad \xi = \frac{\xi_0}{H_{ref}}|\mathbf{H}|.$$

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4. Boundary conditions for the h-lattice

The idea. Two edges of the domain need care. At the **walls**, the field just outside the fluid is the applied field, and the component *along* the wall must match across it — that fixes the slope of A at the wall. At the **seam** where the periodic box wraps around, there is a subtlety: the field is deliberately stronger at one end than the other, so A does *not* come back to the same value after one trip around. It comes back higher by a known amount. We have to teach the periodic wrap about this jump, or the field solver will fight itself at the seam.

4.1 Walls

Tangential $\mathbf{H}$ is continuous and the exterior field is the applied one, so

$$\left.\frac{\partial A}{\partial y}\right|_{y=0,H} = \mu_0\left[H_{x,app}(x,t) + M_{x,w}\right].$$

On the lattice this is specular reflection plus flux injection, with the data evaluated at the point where the population hits the wall, $x_w = i - c_{kx}/2$:

$$h_k(i,j_w) = h_{\bar{k}}^*(i_s,j_w) + 6\epsilon\Delta t\, w_k\,\gamma(x_w), \qquad \gamma_{bot/top} = \mp\mu_0\left[H_{x,app}(x_w) + M_w\right].$$

4.2 The unrolled-ramp closure

The idea. Picture the periodic channel as one turn of an endless spiral ramp, where each turn is a bit higher than the last. The height is A. Going once around, A climbs by a fixed step ΔA. There are actually two contributions to that step: the applied field ramp, and the fact that the magnetization itself is a little stronger at the strong-field end. We track both so the field solver stays exactly consistent and conserves $\int A$.

A_{app} is not periodic in x. The domain is treated as one segment of an infinite ramp:

$$A(x+L,y) = A(x,y) + \Delta A(y,t), \qquad \partial_x A(x+L,y) = \partial_x A(x,y),$$

with **u**, p, ϕ periodic. Since M_x also ramps, the jump has two parts:

$$\Delta A(y) = \mu_0 k L\,(y - y_c) + \mu_0 \int^{y} \Delta M_x \, dy', \qquad \Delta M_x(y) = M_x(x+L,y) - M_x(x,y).$$

ΔM_x is obtained from two ghost columns at $x = -1$ and $x = n_X$ whose magnetization is integrated with the same local ODE, driven by the ghost field $H_x(x \pm L) = H_x(x) \pm kL$, B_y periodic, and by the (periodic) vorticity and phase field of the image column. The two seam crossings are averaged so that $\int A$ is conserved exactly.

4.3 Lattice form of the seam

The idea. In lattice-Boltzmann you do not move field values across the seam, you move little population packets h_k. Each packet carries not just the local A but also a piece of its slope. So when a packet crosses the seam, we add to it exactly the jump in A *and* the jump in the slope of A. Because the ramp is a straight line in y, this correction is exact — not an approximation — which we confirmed to fifteen decimal places.

The populations are near-equilibrium, $h_k \approx w_k\,[A - \tau_h\,\mathbf{c}_k \cdot \nabla A]$ before collision. Collision reduces the gradient term by $(1 - 1/\tau_h)$, so after collision

$$h_k^*(x+L,y) = h_k^*(x,y) + J_k(y), \qquad J_k = w_k\,[\Delta A - (\tau_h - 1)\Delta t\, c_{ky}\, \partial_y \Delta A]\,.$$

Populations entering column 0 from the left have J_k subtracted, those entering column $n_X - 1$ from the right have it added, evaluated at the row they come from. Corner populations get both the wall rule and the jump. Because the applied part of ΔA is linear in y, this relation is exact, not an approximation: for a linear field the Chapman-Enskog form is the exact steady lattice solution, which was verified to 10^{-15}.

4.4 Field recovery

The idea. Once we have A on the lattice, we need **H** back out. The quick way (read it from the packet "moments") is exact for the along-channel component but has a small error in the sideways component, precisely because our field has a mixed second derivative $\partial_x \partial_y A$. We measured that error, showed it only touches a component that does not affect the flow, and kept the quick method. Finite differences of A would be exact if we ever need it.

H is recovered from the first moments of h. This is exact for H_x, but H_y carries a lattice defect of order k whenever $\partial_x \partial_y A \neq 0$, which is exactly this field. Finite differences of A with jump-corrected ghosts are exact instead (Section 7.1). The defect was quantified and shown to be harmless here, so the first-moment recovery was kept.

4.5 Predictor for the oscillating field

The idea. The oscillating field changes every timestep, and the magnetic solver, with only 64 sub-steps, cannot chase a moving target fast enough — it always lands slightly behind. The fix is to give it a head start: before each solve, we shift the whole field by the *exact* amount we know the uniform oscillating part changed, and let the solver clean up only the small remaining correction. This is like handing someone the answer to the easy part of a problem so they can spend their effort on the hard part.

The h-lattice cannot track $H_0 \cos \omega t$ with 64 sub-steps (Section 7.3). Before each sub-cycle, h is shifted by the exact lattice solution of the change in the uniform applied field,

$$h_k \mathrel{+}= w_k \left[\Delta A_{app} - \tau_h \Delta t \, c_{ky} \, \mu_0 \, \Delta H\right], \qquad \Delta H = H_0 \left[\cos \omega t_n - \cos \omega t_{n-1}\right],$$

so the sub-cycle only relaxes the induced part.

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5. Why the flow is parabolic even though $F_y \neq 0$

The idea — this is the question that looks most suspicious, and it has the cleanest answer. We have a force with a sideways (y) component, yet the velocity profile comes out as a clean parabola, exactly as if the sideways force were not there. How? Because a sideways force that depends only on how far you are from the centreline cannot *move* anything — it just gets quietly balanced by pressure. Think of a column of water in a bucket: gravity pulls down hard, but the water does not flow, because a pressure gradient exactly cancels it. Same here: the sideways magnetic force is cancelled by a sideways pressure gradient, and the flow never feels it. The only force that can drive flow down the channel is one that varies *along* the channel — and only its average does the pushing.

The mathematics. Take a steady, fully developed, incompressible flow with $\mathbf{u} = (u(y), 0)$. Continuity is satisfied identically. The two momentum components decouple:

$$x: \quad 0 = -\frac{\partial p}{\partial x} + \eta \, u''(y) + F_x, \qquad\qquad y: \quad 0 = -\frac{\partial p}{\partial y} + F_y(y).$$

The y-equation contains **no velocity at all**. Whatever $F_y(y)$ is, it is absorbed by a transverse pressure distribution

$$p(x, y) = -Gx + \int^y F_y(y') \, dy' + \text{const},$$

and the flow never sees it. A transverse force can only drive motion if it has a non-zero curl that survives in the x-equation, i.e. if it varies with x; a force that depends on y alone is exactly balanced by pressure. In this problem

$$F_{K,y} = \mu_0 \left(M_x \partial_x H_y + M_y \partial_y H_y\right) = -\mu_0 k M_y \approx \mu_0 k^2 M \frac{y - y_c}{H_d} = O(k^2),$$

which is both tiny and a function of y only, so it changes $p(y)$ by parts in 10^4 of the axial pressure drop and changes u by nothing.

Three refinements matter for the real simulation:

1. F_x **does vary with** x, because M_x follows the local field $H_d + k(x - x_c)$. Write $F_x = \bar{F}_x + F'_x(x)$. In a periodic channel, only $\bar{F}_x$ drives flow; the fluctuating part is balanced by a periodic pressure modulation. Measured x-variation of u: 4×10^{-5} relative at $r = 0.3$, against a $\pm 2.5\%$ variation of the force.
2. **The Kelvin force is a gradient for aligned M.** With $\mathbf{F}_K = \nabla\Phi$, $\Phi = \mu_0 \int_0^{|\mathbf{H}|} M\, dH$, a periodic field would give zero net drive. The drive is exactly the potential difference across one period, $\bar{F}_x = [\Phi(L) - \Phi(0)]/L = \mu_0 k \langle M_x \rangle$, which is why the field must be non-periodic and the lattice needs the seam jump. (This is the deep reason the field cannot be periodic: a force that is a pure gradient can only pump if the "potential" is higher at one end than the other.)
3. **The oscillating part of** F_x produces a plug oscillation $U_{osc} \sim F_{osc}/(\rho\omega)$, which superposes linearly on the mean and averages to zero. Measured: snapshot $u_{max} = 1.5376 \times 10^{-3}$ versus time-averaged 1.5419×10^{-3}, a difference of 0.3%.

So the time-averaged profile is parabolic to the precision we can measure, and the whole measurement reduces to one number per run: the curvature u''.

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6. How the viscosity is measured

The idea. A parabolic profile has one number that matters: how sharply it curves. That curvature is set by the balance between the push and the viscosity — a stiffer fluid curves less for the same push. So if we know the push and measure the curvature, we can read off the effective viscosity, and from that the rotational-viscosity factor g that Bacri's theory predicts. To be safe we always run a twin experiment (a "control") that uses a plain mechanical push instead of the magnetic one, which lets us catch any extra magnetic force masquerading as a viscosity change.

6.1 Definition

Rotational viscosity in Bacri's normalization:

$$\overline{\mu_0(\mathbf{M} \times \mathbf{H})_z} = -6\eta_h \phi_p\, g\, \Omega, \qquad \eta_{eff} = \eta_h \left(1 + \tfrac{3}{2}\phi_p\, g\right).$$

If the torque is proportional to the local vorticity, then $F_{rot} = \partial_y(\Delta\eta\, \partial_y u)$ and the momentum balance collapses to

$$\eta_{eff}\, u'' = -F_K.$$

The solver writes the time-averaged profiles of u, F_K and F_{rot} and fits η_{eff} by least squares over interior rows, skipping the two rows adjacent to each wall where the halfway-wall stencil distorts u''.

6.2 The independent reference

The idea. Alongside the full simulation we run a stripped-down, single-point calculation of the same magnetization physics — no flow, no lattice, just the little relaxation equation at one imagined point. This gives us an independent prediction of g to compare against. It reproduces the exact textbook answer in the static case to six digits, so we trust it as a ruler.

`mrsh_reference` integrates the same ODE at a single point with $H_x(t) = H_d + H_0 \cos \omega t$, $H_y = -M_y$ (the planar demagnetizing field) and a small imposed Ω, then reports $\langle M_x \rangle$ (which sets k for the target drive) and g. It reproduces the closed-form static value $g_{DC} = \xi L^2/(\xi - L)$ to six digits, and it is the yardstick for every run.

6.3 The extra magnetic body force

The idea — this is the new physics. Remember the forced sideways field H_y from Section 2.1. In a steady field the magnetization lines up perfectly and $\mathbf{M} \times \mathbf{H}$ is zero, so that sideways field does nothing. But in an *oscillating* field the magnetization lags, $\mathbf{M}$ and $\mathbf{H}$ are no longer parallel, and their cross product is non-zero. Worse, it grows with distance from the centreline, and the curl of something that grows linearly is a *constant* — a uniform extra push. In a parabolic flow, a uniform extra push and a change in viscosity bend the profile in exactly the same way, so the naive fit blames the viscosity for a force. We compute this extra push independently (at zero flow-rotation, to prove it is a force and not a viscosity) and correct for it.

$\langle \mu_0(\mathbf{M} \times \mathbf{H})_z \rangle$ acquires a part proportional to $k(y - y_c)$, whose curl is a **uniform** force:

$$F_{extra} = \tfrac{1}{2}\, \partial_y \big\langle \mu_0(\mathbf{M} \times \mathbf{H})_z \big\rangle \Big|_{\Omega=0} \equiv \varepsilon\, G.$$

It is computed at zero vorticity, so it is a body force, not a viscosity. In Poiseuille flow u'' is constant, so a uniform force and a viscosity change are algebraically indistinguishable: the fit returns $\eta_{meas} = \eta_{true}/(1+\varepsilon)$, and the correction is $\eta_{true} = \eta_{meas}(1+\varepsilon)$.

6.4 Why every case has a control

The idea. Because of 6.3, we cannot trust the magnetic-drive run by itself — it hides that extra force inside the viscosity. So each magnetic run has a twin that is pushed mechanically with the same average force but no gradient, hence no sideways field and no oscillating push. The twin measures g cleanly; the difference between the twins is exactly the magnetic pumping physics we are after.

Because of 6.3, each Kelvin-driven case is paired with a control: identical field, same mean drive, but supplied as a constant body force with $k = 0$. The control has no transverse field and no oscillating force, so it measures g directly. The difference between the pair isolates the magnetic pumping terms.

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7. Tests and trials, in order

The idea. We did not trust anything until it survived a test we already knew the answer to. The sequence below builds confidence from the ground up: first the field alone (no fluid), then the field tracking in time, then the fluid coupling, then the full sweep — and finally the two episodes where the numbers looked wrong and we chased down why. Two of those chases found real physics; one found a bug in our own analysis. All three are kept here, because knowing *why* a control run is mandatory is part of the result.

7.1 Field exactness with no fluid response ($M_s = 0$)

50x50, NSUB_H $= 64$, $k = 1.21 \times 10^{-5}$ and 10^{-3}:

quantity	result
$\max\|H_x - H_{x,app}\|$	4.2×10^{-15} (all nodes, incl. seam and corners)
$\max\|\partial_x H_x - k\|$	9.5×10^{-16}
A_{resid}	0
$\max\|H_y - H_{y,app}\|$, first moment	$1.71\,k$
$\max\|H_y - H_{y,app}\|$, FD of A	1.1×10^{-14}
$\max\|\partial_y H_y + k\|$, first moment	$0.49\,k$

The H_y defect appears only when $\partial_x \partial_y A \neq 0$. It was retained because it enters only $F_{K,y}$ (pressure, Section 5) and because **M** aligns with the same computed **H**, so the torque is unaffected. Confirmed by 7.5.

A separate Python replica of the h-lattice (16x12, $k = 0.05$, $M = 0$) converged to A_{app} within 10^{-15} from a randomised initial state, validating the jump rule, the wall rule and the corner handling independently of the C++ code.

7.2 Tracking of the oscillating field

Replica test with $M = 0$, 50x50, $\epsilon = 9.2$, 64 sub-steps: channel-mean amplitude error $\leq 5 \times 10^{-4}$, phase lag 0.013 steps, and a non-uniform (quadrature) component of 6.5×10^{-4}, 9.8×10^{-3}, 1.3×10^{-2} of H_0 at $\omega\tau_B = 0.1,\ 1.5,\ 2$. An attempt to speed convergence by lowering ϵ to 4.6 made it worse (3.5%), refuting a telegraph-equation estimate I had made.

7.3 Sub-cycle convergence, and the predictor

Why it matters: this is the test that justified the predictor of Section 4.5, and it showed that 64 sub-steps alone were *not* good enough for the oscillating case.

At $n_Y = 16$, $\xi_0 = 20$, $\omega\tau_B = 1.5$, $k = 0$ with a body force:

configuration	H_max_delta	g
64 sub-steps, no predictor	1.36%	0.0907
256 sub-steps, no predictor	0.023%	0.0733
64 sub-steps, predictor	0.039%	0.0738
256 sub-steps, predictor	6×10^{-6}	0.0736

So 64 sub-steps alone were not converged, with an error of about ± 0.02 in g whose sign depended on n_Y. With the predictor, 64 sub-steps are converged to 2×10^{-4}. This also casts doubt on the earlier Fig. 3 agreement, which used 64 sub-steps without a predictor.

7.4 Regression against the validated solver

With $k = 0$, $H_d = 0$, $H_0 = 1.73$ and the body force restored, the new code reproduces the original Fig. 3 solver exactly across ux, Hx, Hy, Mx, My, Fm_x (identical to CSV precision). All new machinery is therefore inert when the gradient is switched off. (This is the reassuring test: with the new field turned off, we recover the old, already-validated code bit for bit.)

7.5 Static-field equivalence: Kelvin drive versus body force

50x26, NSUB_H $= 8$, 20k steps, $H_0 = 0$. Kelvin drive with k chosen for $G = 2 \times 10^{-7}$, against the original solver with a uniform static field and $F_b = 2 \times 10^{-7}$:

$$\max|u_{new} - u_{ref}|/u_{max} = 1.4 \times 10^{-7}, \qquad \text{both versus theory: } 3.8 \times 10^{-4}, \qquad |u_y| \leq 1.5 \times 10^{-9}.$$

In words: a magnetic pull and a mechanical push of the same strength produce the same flow to seven digits, confirming the Kelvin drive is implemented correctly.

7.6 The seam, before and after the magnetization jump

Before: the force at the seam columns was 1% high and at columns 1-2 about 0.1% low, with zero mean; the error scales as $\chi_d L/2$, which is 1.3% at $\tau_B = 20$ but $O(1)$ near the zero crossings of an AC field. After the ghost columns and the ΔM_x term: the column-mean force varies smoothly and linearly across the domain, matching the physical ± 2.5% from $kL/H_d = 5$%, $\langle M_x \rangle$ matches the reference to 10^{-4}, and the x-variation of u is 4×10^{-5}.

7.7 The main sweep

36 Kelvin runs and 36 controls: $\xi_0 \in \{1, 2.5, 5, 10, 15, 20\}$, $\omega\tau_B \in \{0, 0.1, 0.5, 1, 1.4, 2\}$, $H_0 = 1.73$, $H_d/H_0 = 0.3$, 120k steps on 50x50, roughly 18 hours.

Static points. At $\omega\tau_B = 0$ the field is static at $H_d + H_0 = 2.249$, so $\xi_{local} = \xi_0 \cdot 2.249/1.73$:

ξ_0	ξ_{local}	g analytic	g simulated
1	1.300	0.2190	0.2189
2.5	3.250	0.6151	0.6149
5	6.500	0.8231	0.8230
10	13.000	0.9172	0.9171
15	19.500	0.9461	0.9460
20	26.000	0.9601	0.9599

The simulation hits the exact closed-form answer to four digits — the strongest single check in the project.

Controls. All 36 match the reference to $\leq 4 \times 10^{-4}$, including negative g.

Kelvin at $\omega\tau_B = 0$. Identical to the control to four decimals: a steady Kelvin force behaves exactly like a body force, so all discrepancies are dynamic.

Viscosity curve ($\xi_0 = 20$, controls): η_{eff}/η_h = 1.288, 1.241, 1.176, 1.096, 1.034, 0.949 as $\omega\tau_B$ goes 0 to 2. The last point is 5% below the zero-field viscosity: **negative rotational viscosity, measured from the flow rate** — the fluid flows faster than it would with no field at all.

7.8 Hypotheses tested for the Kelvin-control offset

The story. The magnetic runs disagreed with their twins by up to -0.47 in g — a huge gap. We proposed four explanations and killed three before finding the real one.

1. **Extraction artifact.** I first estimated the Kelvin force by subtracting the control's total force, which also removed the rotational force and inflated the drive by 3%. Wrong, and corrected by writing F_K to file separately.
2. **Stokes-layer streaming.** Ruled out: the oscillating flow is only 0.3% of the mean (snapshot versus time average), far too small to rectify into 25% of the drive.
3. **Quadratic rectification.** Ruled out by an amplitude scan at $\xi_0 = 10$, $\omega\tau_B = 2$: $\Delta g = -0.4456,\ -0.4457,\ -0.4457$ for $G = 2 \times 10^{-7},\ 1 \times 10^{-7},\ 5 \times 10^{-8}$. Amplitude-independent to four decimals, so nothing quadratic in the drive.
4. **Transverse-field stress.** Confirmed, see 7.9.

7.9 The bias sweep, and the confirmation

The idea. The winning explanation was the sideways field of Section 6.3. To prove it, we varied the steady bias $r = H_d/H_0$: more bias means the field reverses less, the magnetization lags less, and the extra force should shrink in a way we can predict from the single-point model. It matched over a 17-fold range.

$\xi_0 = 10$, $\omega\tau_B = 2$, varying $r = H_d/H_0$. Measured ε from $\eta_{meas}/\eta_{true} = 1/(1+\varepsilon)$; predicted ε from the zero-vorticity single-point calculation of 6.3:

r	k	ε measured	ε predicted	ratio
0.3	9.36×10^{-5}	0.1631	0.1751	1.074
0.5	5.17×10^{-5}	0.0913	0.0974	1.067
0.8	3.27×10^{-5}	0.0253	0.0272	1.075
1.0	2.85×10^{-5}	0.0096	0.0105	1.094

A 17x range reproduced with a constant 7% systematic. The predicted ε is identical for $Y_0 = 4,\ 8,\ 12.5$, confirming that the force is uniform across the channel. This explains every feature: ε is independent of G (both F_{extra} and G scale as k); $\varepsilon = 0$ at $\omega\tau_B = 0$ (no lag, so $\mathbf{M} \times \mathbf{H} = 0$); ε grows with $\omega\tau_B$ (more lag) and falls with r (less reversal).

7.10 The low-frequency anomaly: an averaging bug

The story. At the lowest frequency the correction failed, and the failure smelled wrong: it did not scale with the drive, it flipped sign when we changed the grid, and the control runs never showed it. Those three clues together point away from physics and toward the analysis. The culprit was our time-averaging window: it did not cover a whole number of oscillation periods, so a sliver of the oscillating force leaked into what we called the "mean". Rounding the window to whole periods fixed it.

At $\omega\tau_B = 0.1$ the correction failed, needing $\varepsilon \approx -0.03$ where the model gave $+0.0002$. Diagnosis:

- amplitude scan: $g = 0.7167$ at both $G = 2 \times 10^{-7}$ and 5×10^{-8}, so not rectification;
- the sign flipped between $n_Y = 50$ (+0.125 in g) and $n_Y = 26$ (-0.086);
- the controls, which have no oscillating force, showed nothing.

Cause: the averaging window was 40 field periods capped at `nSteps/4`, which made it a non-integer number of periods, so the oscillating Kelvin force leaked into the mean: leak/$F_{osc} \leq 2/(\omega M)$, which is 4% of $F_{osc} \approx G$ at $\omega\tau_B = 0.1$ and 0.2% at $\omega\tau_B = 2$. Rounding the window to whole periods fixed it:

averaging window	g	g_{ref}	residual
40 periods, capped	+0.7189	+0.8028	-0.084
whole periods	+0.8114	+0.8028	$+0.009$

Rerunning the $\omega\tau_B = 0.1$ column at full resolution gave residuals of $+0.0008$ to $+0.0037$, a 30-50x improvement, with $\varepsilon \leq 0.0004$ there.

7.11 Final accuracy

After the fix and the ε correction, $g_{corr} - g_{ref}$ across all 30 AC cases:

ξ_0 $\omega\tau_B$	0.1	0.5	1.0	1.4	2.0
1	+0.001	+0.000	+0.001	+0.001	+0.001
2.5	+0.003	$-$0.000	+0.003	+0.003	+0.003
5	+0.004	$-$0.001	+0.004	+0.006	+0.008
10	+0.003	$-$0.001	+0.002	+0.003	+0.008
15	+0.003	$-$0.001	+0.002	+0.001	+0.005
20	+0.003	$-$0.001	+0.002	+0.000	+0.003

Worst case 0.008 over a range of g from $+0.96$ to -0.62. The largest residuals coincide with the largest ε, consistent with its 7% systematic.

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8. Demagnetization

The idea. A slab of magnetized fluid makes its own field, pointing back against the magnetization — the "demagnetizing field". Our flat channel has a specific amount of it; Bacri's thin cylindrical capillary has a different amount. So even with everything else identical, our g is shifted from his by a known, calculable amount that scales with how strongly magnetized the fluid is ($\lambda = M_s/H_{ref}$). We have to subtract this shift before comparing to his figures. Reassuringly, it vanishes exactly in the static case, which is why our static checks were so clean.

The planar channel has $H_y = -M_y$ inside (normal B continuous, no exterior normal field), which Bacri's capillary geometry does not. Reference calculations at $\xi_0 = 20$, $\omega\tau_B = 1.5$:

τ_B	$\lambda = M_s/H_{ref}$	g with demag	g without	Δ	Δ/λ
20	0.01002	0.07031	0.04786	+0.02245	2.24
80	0.00251	0.05385	0.04791	+0.00594	2.37
320	0.00063	0.04942	0.04791	+0.00151	2.41

$$g = g_\infty(\xi_0, \omega\tau_B) + c\,\lambda, \qquad c \approx 2.3, \qquad \lambda = \frac{3\eta_h \phi_p \xi_0}{\mu_0 \tau_B H_{ref}^2}.$$

Alternatives were excluded: refining the time step 4x at fixed τ_B changes g by only +0.001, and g without demagnetization is τ_B-independent to four digits. At DC the effect cancels exactly, because the demagnetizing field enters both the restoring torque and the measured torque by the same factor. Consequences: a τ_B sweep at fixed $\omega\tau_B$ is **not** a null test, since it varies λ; and comparisons with Bacri need either $\lambda \to 0$ extrapolation or the demagnetization-free reference, which the solver prints as `g_ref_nodemag`.

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9. Operating limits

The idea. Every knob has a safe range, and the pre-flight check refuses to run outside it. The ramp must be gentle (or the field varies too much across the box), the timestep small enough to resolve the oscillation and the stiff magnetization relaxation, the flow slow enough to stay in the linear-response regime where g is even defined, and the averaging window a whole number of periods (Section 7.10 taught us that one the hard way).

limit	expression	reason
field ramp	$kL/H_d = 3.33 \times 10^{-4}\,\tau_B/(\xi_0 L(\xi_0)) \le 0.02$	local ξ spread, seam closure
RK4 stability	$\xi_0/(8\tau_B) < 2.785$	stiffest MRSh rate is $\sim \xi_0/\tau_B$
linear response	$\Omega_w\tau_B \approx 5 \times 10^{-5}\tau_B < 0.05$	g defined at small $\Omega\tau_B$
time resolution	$\omega\Delta t = \omega\tau_B/\tau_B \le 0.1$	**H** updated once per step
seam closure	$\chi_d L/2 \le 25\%/\tau_B$	residual local force error
averaging	whole number of field periods	Section 7.10

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10. Results worth reporting

1. **Negative rotational viscosity in a magnetically driven flow.** η_{eff}/η_h falls from 1.288 to 0.949 across $\omega\tau_B \in [0, 2]$ at $\xi_0 = 20$, measured from the flow rate rather than inferred from a torque model. *(In plain terms: the fluid, pushed by nothing but a magnetic field, flows faster at high oscillation frequency than the same fluid with no field — the field has made it effectively thinner.)*
2. **Two magnetic pumping mechanisms.** A graded AC field drives flow through the Kelvin force $\mu_0 k\langle M_x\rangle$ and through the antisymmetric-stress force $\frac{1}{2}\nabla\times\langle\mu_0\mathbf{M}\times\mathbf{H}\rangle$ that arises from the transverse field Maxwell requires. The second reaches 18% of the first at $\omega\tau_B = 2$, $r = 0.3$; it vanishes in a static field, is independent of the drive amplitude, is uniform across the channel, and is predicted quantitatively by a zero-vorticity single-point calculation. *(In plain terms: there are two separate magnetic pushes, not one — the obvious pull-toward-strong-field, and a subtler push from the lagging magnetization being twisted by the field's forced sideways tilt. The second is new and is easy to mistake for a viscosity change.)*
3. **A planar demagnetization shift** of $+2.3\,\lambda$ in g, absent from Bacri's capillary theory, which must be removed before any comparison with his figures.

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11. Open items

1. The 7% systematic in the predicted ε: likely the one-step lag in the demagnetizing field or the y-variation of M_x in the reference.

2. The zero-mean-field ($H_d = 0$, body-force) runs needed for a direct Bacri comparison.
3. Mesh independence at $n_Y = 50$, outstanding from the previous phase.
4. Re-verification of the earlier Fig. 3 results, which used 64 sub-steps without the predictor and therefore carry a possible ± 0.02 in g, partly cancelled by the demagnetization shift of opposite sign.
5. The co-oscillating gradient, $k \to k \cos \omega t$, as the design that suppresses the transverse-field stress: there $H_y \propto \cos \omega t$ and the mean of the cross-term changes character.

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12. Files and reproduction

file	role
`phaseField_poiseuille_linearField_AC.cpp`	solver: linear field, seam jump, ghost columns, predictor, reference, `Fk_x` output
`sweep.py`	patch (command-line args, $\omega = 0$ guard, time averaging, in-code η_{eff} fit), smoke test, pre-flight, sweep, collect
`run_sweep.sh`	toolchain check and entry point
`correct_kelvin.py`	computes ε and adds `g_corr` to `summary.csv`
`plot_profiles.py`	velocity and force profiles

```
./run_sweep.sh --smoke     # patch, build, three validation cases, pre-flight table
./run_sweep.sh             # full sweep, then summary.csv
python3 correct_kelvin.py summary.csv
python3 plot_profiles.py runs/<tag>_kelvin runs/<tag>_ctrl --forces -o fig.png
```

The smoke test checks the patch anchors, compilation, the exit code, a `RESULT` line, the absence of NaN, the row count of `profile_avg.csv`, $k > 0$, and that the static reference matches $g_{DC} = \xi L^2/(\xi - L)$ to 10^{-3}. If any check fails, nothing else runs.